BC Calculus II/III
Fall 2024

Instructor: Dr. Anderson Trimm
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Office hours: I am available daily during S&E.

Availability outside of class: All students are welcome to meet me during the office hours listed above without appointment. If you need to meet with me outside of this time, please send me an email to make an appointment in order to ensure my availability.

Class Meeting Information: Mod 1 in A155 on ABCD days.

Attendance: Please refer to the attendance policies in the IMSA 2024–2025 Student Handbook. My expectation is that you come to class prepared every day and are an active participant in all group activities and class discussions. Please be seated with your notebook out by the beginning of each class.

Text/Materials: Text: Strang, G. and Herman, E. *Calculus, Vols 1 and 2*, 2016.
Graphing Calculator (TI-89 or TI-84 recommended).
Three-ring binder for class notes, worksheets, quizzes, etc.
Laptop computer
You should bring your binder, calculator, and laptop to class daily.

Canvas: I will post copies of all course handouts, assignments and due dates, notifications of upcoming assessments, and so forth on Canvas. Please make sure to **turn notifications on**.

Course Description

BC II/III is the second semester of the two-semester BC Calculus sequence. Topics of study include improper integrals, differential equations, sequences, series, and parametric equations.

Course Schedule

Here are the major divisions of this course:

- **Techniques of Integration**
- **Applications of Integration**
- **Improper Integrals**
- **Differential Equations**
- **Sequences and Series**
- **Power Series**
- **Vectors and Parametric Equations**

Learning Standards

Integration Techniques, Part II (7, 3 core)

- I.6: I can evaluate integrals by trig substitution.
- I.7: I can evaluate integrals by partial fractions.
- I.8: (Core) I can evaluate a definite integral as the limit of a Riemann sum.
- I.9: (Core) I can approximate a definite integral using the left-endpoint, right-endpoint, midpoint, trapezoidal, and Simpson's rules.
- I.10: I can determine the maximum absolute error in the approximations in 1.9, and determine the number of terms in the Riemann sum needed to keep the maximum error under a given threshold.
- I.11: (Core) I can calculate improper integrals using the definition.
- I.12: I can determine whether a given improper integral converges using the comparison test and the limit comparison tests.

Applications of Integration (6, 3 core)

- IA.1: (Core) I can find the area between two curves.
- IA.2: (Core) I can find the volume of a solid of revolution using the disk method.
- IA.3: I can find the volume of a solid of revolution with a hollow cavity using the washer method.
- IA.4: I can find the volume of a solid of revolution with a hollow cavity using the method of cylindrical shells.

- IA.5: I can find the surface area of a solid of revolution.
- IA.6: (Core) I can find the arc length of a curve.

Differential Equations (4, 1 core)

- DE.1: I can find a slope field and use it to sketch an approximate solution curve.
- DE.2: I can use Euler's method to find a piecewise linear approximation to the solution of a differential equation.
- DE.3: (Core) I can solve a separable differential equation.
- DE.4: I can use differential equations to model exponential and logistic growth.

Sequences and Series (6, 4 core)

- S.1: I can determine when a given geometric or telescoping series converges and I can find the sum. I can use geometric series to convert a repeating decimal to a ratio of integers.
- S.2: (Core) I can use the nth term Test for Divergence to prove that a divergent sequence diverges.
- S.3: (Core) I can use the Integral Test to determine whether a given series converges or diverges.
- S.4: (Core) I can use the Alternating Series Test to determine whether a given alternating series converges.
- S.5: I can use the Alternating Series Error Theorem to estimate an alternating series to any desired accuracy.
- S.6: (Core) I can use the Ratio Test to determine whether a given series converges absolutely.

Power Series (5, 3 core)

- PS.1: (Core) I can use the Ratio Test to find the interval of convergence of a given power series
- PS.2: I can write a given function as a power series by manipulating the geometric series formula. I can differentiate and integrate a power series.
- PS.3: (Core) I can write the Maclaurin series of a given function by using a known Macluarin series and performing appropriate operations on series (multiplying by a constant, integrating, or differentiating).
- PS.4: (Core) I can approximate an analytic function by a Taylor polynomial within a specified error.
- PS.5: I can write the Taylor series of a function using the definition of a Taylor series.

Grading

Types of Graded Assignments

1. **WebWork**: WebWork problem sets will be assigned frequently to reinforce learning in class. These mostly focus on computational techniques.

2. **Problems of the Week**: These problems will be assigned approximately weekly. These assignments go beyond basic calculations by diving deeper into the theory, exploring applications, or integrating multiple concepts. It is strongly preferred that solutions are typeset in LaTeX, but this is not required as long as they are handwritten neatly.

3. **Checkpoints**: We will have frequent checkpoints throughout the course to assess the 28 learning standards listed above. Each checkpoint will cover 1-4 learning targets, which will be announced in advance.

4. **Final Exam**: There will be a cumulative final exam covering the 28 learning targets listed above.

How Assignments are Graded

The assignments listed above will be graded as follows.

1. **WebWork**: Each WebWork assignment is worth some specified number of points. WebWork grades these automatically. Unlimited attempts are allowed.

2. **Problems of the Week**: These will be graded for accuracy out of 10 points.

3. **Checkpoints**: Each learning target assessed will be given one of the following scores out of 4 possible points:

   4 : Mastered
   3 : Nearing Mastery
   2 : Proficient
   1 : Nearing Proficiency
   0 : Beginning

   Criteria for each score will be communicated in the directions for each assignment.

4. **Final Exam**: There will be a cumulative final exam covering the 28 learning targets listed above. The questions will be at the same level of difficulty as those on the checkpoints. The final exam grade will be recorded as a percentage. More details will be provided closer to the time of the final exam.

Assignment of Letter Grades

Final letter grades will be assigned according to the following table:

Letter Grade	All core met at 2 or higher	Total Score	Sum of Scores (≥ 2)	Sum of Scores (≥ 3)	Sum of Scores (4)	POTWs	WebWork
A	Yes	≥ 98	-	-	≥ 72	≥ 85%	≥ 90%
B	Yes	≥ 84	-	≥ 49	-	≥ 75%	≥ 80%
C	Yes	≥ 70	≥ 42	≥ 14	-	≥ 60%	≥ 70%
C-	Yes	≥ 56	≥ 28	≥ 7	-	≥ 50%	≥ 60%

- *All* requirements in a given row must be met in order to receive the corresponding grade. If all requirements for a grade of C- are not met, then a grade of D will be assigned.
- If all but one of the requirements in the next row are met *or* the final exam score is 85% or above, then a final letter grade of *C* or *B* will be augmented with a '+'.
- If the final exam score is below 50%, then a final letter grade of *C*, *B*, or *A* will be augmented with a '−', while a final letter grade of *C*+ or *B*+ will be lowered to *C* or *B*, respectively.

Collaboration / Use of Outside Resources and Late Work / Retake Policy

- **WebWork Assignments:** No collaboration is permitted. Unlimited responses will be allowed, so you may keep trying a problem if you do not get it right initially. Talk to me if you get stuck; do not mindlessly try many answers. Calculators (online or physical) may be used for tedious number-crunching, but may not be used to do any essential calculation. For instance, if the point of the assignment is to practice an integration technique, you may not use integral-calculator.com, or any similar resource, to evaluate an integral. As these assignments must be completed in a timely manner to support your day-to-day learning in class, extensions will only be granted due to extenuating circumstances. *These assignments may not be retaken or revised.*
- **Problems of the Week:** These will be completed with a partner, and one assignment will be turned in together. Make sure both names are on the submission. It is expected that both partners make an equal contribution to each assignment. If you get stuck, you may consult with me, other students, teachers, friends, parents, etc., but your final work must be your own. You may ***not*** search the web or use AI for hints; come ask me instead. Opportunities to retake or revise Problems of the Week will be announced in class. Any such retakes or revisions will generally be due one week after the assignment is returned. Please staple your

revision or solution to your original submission; do not edit the original. *Any solution turned in which was simply copied from the internet, another group, or was generated by AI, will receive a grade of 0 and will not be able to be revised or retaken.*

- **Checkpoints:** These will be taken individually in class, mostly paper and pencil, but sometimes on computer. Calculators will be permitted on some of these. There will be one retake for each checkpoint, approximately one week after the original checkpoint is returned. The final exam will also function as a retake for all learning targets. The retakes will cover the same learning targets at the same level of difficulty, but will consist of different questions.

Electronics Policy

All phones and other electronic devices should be silenced and put away before the beginning of class. Laptops should remain put away during class, unless you are directed to use them for an assignment or activity.

Use of AI / Internet Policy

I will try my best to provide any needed resources, so please ask if you need something. Note that videos and internet resources that I have not vetted, or responses generated by AI, may contain errors, lead to misconceptions, or be just plain wrong. For this reason I strongly discourage using any such resources without running them by me first.

Teaching and Learning Philosophy

My approach to teaching the course will focus on developing conceptual understanding, procedural fluency, mathematical reasoning, and problem-solving. All students are strongly encouraged to contribute to class discussions. I encourage everyone to challenge themselves and to step outside of their comfort zone.

Diversity, Equity, and Inclusion Statement

Everyone has the right to be included and treated with the highest level of respect, to be addressed by their preferred name and pronouns, and to have their identity respected. Students are encouraged to provide the instructor with regular input on how they are experiencing the course throughout the semester.

Equity Goals

- Ensure that each student has safe and accessible opportunities and support to be able to enhance their knowledge and understanding of mathematics.
- Provide multiple approaches for learning and assessment to meet the various needs of our diverse group of students and their backgrounds.
- Helping students see the value in developing the ability to have mathematical discussions and to disagree cooperatively in critiquing the reasoning of others, and not adversarially.

- Ensuring that all students attain mathematics proficiency and increasing the numbers of students from all racial, ethnic, linguistic, gender, and socioeconomic groups who attain the highest levels of mathematics achievement. (NCTM 2020)
- Creating, supporting, and sustaining a culture of access and equity that is responsive to students' backgrounds, experiences, cultural perspectives, traditions, and knowledge. (NCTM 2020)

Academic Behavior Code

Students are expected to follow the IMSA 2024–2025 Student Handbook regarding academic honesty and IMSA's Academic Behavior Code.

Learning Standards SSLs

Developing the Tools of Thought
I. IMSA students develop automaticity in skills, concepts, and processes that support and enable complex thought.
II. IMSA students construct questions which further understanding, forge connections, and deepen meaning.
IV. IMSA students evaluate the soundness and relevance of information and reasoning.

Extending and Integrating Thought
II. IMSA students recognize, pursue, and explain substantive connections within and among areas of knowledge.

Expressing and Evaluating Constructs
I. IMSA students construct and support judgments based upon evidence.
II. IMSA students write and speak with power, economy, and elegance.

This syllabus is subject to change throughout the semester as needed at the instructor's discretion.